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#include <stdio.h>

int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1, mid;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (arr[mid] == key)
            return mid;        // Element found
        else if (arr[mid] < key)
            low = mid + 1;      // Search right half
        else
            high = mid - 1;     // Search left half
    }
    return -1; // Element not found
}

int main()
{
    int n, key, result;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d sorted elements:\n", n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    result = binarySearch(arr, n, key);

    if (result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");

    return 0;
}

```